PART I: FOUNDATIONS

Topic 3: Visualizing Time Series — The Line Plot

The Core Concept

The **line plot** is the gold standard for time series visualization. It connects data points with a line to emphasize **movement and direction**, making patterns instantly visible to the human eye.

Why It Matters

The human brain struggles to find patterns in a table of numbers, but it is expert at reading shapes. A line plot instantly reveals trends, seasonal waves, and outliers.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

For Precinct North, the table shows: 12, 10, 14, 15, 18, 22, 25... A line plot of these numbers would show a clear "mountain" shape — easy to interpret at a glance.

Common Visualization Mistakes

- **Using bar charts:** Bars emphasize individual values rather than the flow between them. They can obscure trends.
- **Hiding zero:** If the Y-axis does not start at zero, a small change can look like a massive crisis. Always check the Y-axis scale before drawing conclusions.
- **Plotting too many series:** More than 3–4 lines on one chart becomes difficult to read. Use separate panels if needed.

💡 **The Insight:** A well-designed line plot communicates more in 5 seconds than a data table communicates in 5 minutes.

Key Takeaway: Always visualize your data before analyzing it — the shape often tells you where to look.